Assignment 2

② $P(D) = 0.05$

Expected $= \frac{1}{0.05} = 20$

③ $1 - P(D) = 0.95$

$P(6th\ D) = \cancel{0.95^5\ 6\ 0}$

$= (0.95)^5 (0.05)$

$= 0.039$ (3d.p.)

④ $P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}$ where $n = 6$

② $x = 2$

$= \binom{6}{2} (0.05)^2 (0.95)^4$

$= 15(0.00203...)$

$= 0.031$ (3d.p.)

⑤ No, since geometric ~~[illegible]~~ can only ~~[illegible]~~ calculate trials til first success, whereas binomial calculates only number of successes. So binomial can be used for both

⑥ $p(N) = 0.95$

$$\binom{10}{0}(0.05)^0(0.95)^{10}$$

$= 0.599$ (3d.p.)

⑦ $1-\left[\binom{10}{0}(0.05)^0(0.95)^{10}\right]$ I don't want to do $1-0.599$ ~~the~~ I want to make sure there is no rounding error. hahaha num its the same answer

$= 0.401$ (3d.p.)

⑧ No, see ⑤

⑨ $p(D) = 0.01$
$n = 30$

$$P(x=0) = \binom{30}{0}(0.01)^0(0.99)^{30}$$

$= 0.740$ (3d.p.)

⑩ $\lambda = \frac{1}{100}(30) = 0.30$

$$P(x=0) = \frac{e^{-0.3}(0.3)^0}{0!} = 0.741 \text{ (3d.p.)}$$

⑪ $\lambda = \frac{15}{100}(30) = 4.5$ | $P(x \geq 2) = 1 - P(x=0) - P(x=1)$

$$= 1 - (e^{-4.5}) - \left(\frac{e^{-4.5}(4.5)^1}{1!}\right) = 0.939 \text{ (3d.p)}$$

(12)

$$P(X \geq 2) = 1 - \left[\binom{30}{0}(0.15)^0(0.85)^{30}\right] - \left[\binom{30}{1}(0.15)^1(0.85)^{29}\right]$$

$$= 0.952 \text{ (3d.p.)}$$

(13) poisson less accurate, let A = 2 or more earthquakes

B = decreasing winds

$P(A) = 0.952$

$P(B) = 0.740$ indep. $\therefore P(A \cap B) = P(A)P(B)$

$$P(A \cap B) = 0.952(0.740) = 0.704 \text{ (3d.p.)}$$

(14) $A \rightarrow 1 - F_X(X \leq 200) = \sum_{i=0}^{200} \binom{320}{i} 0.65^i (1-0.65)^{320-i}$

$$= 0.811$$

$B \rightarrow 1 - F_X(X \leq 200) = \sum_{i=0}^{200} \binom{230}{i} 0.90^i (1-0.90)^{230-i}$

$$= 0.920$$ $\therefore$ B is better

(15) same eqns, but replace 200 with 210

$A = 0.387$ ←

$B = 0.224$

(16) over 1 year

$A) = 0.65(320)(365) = 75720$

$B) = 0.90(230)(365) = 75555$

A is better over 1 year.